UNREAL ENGINE 5 CORE TECHNICAL GUIDE

Airport Dynasty Pro

Next-Gen Airport Simulation & Global Aviation Architecture (*Matrix Awakens* Tech Stack)

Airport Dynasty Pro combines hyper-realistic global aviation simulation with high-density airport crowd AI and dynamic terminal management. This guide outlines the implementation of Unreal Engine 5 features—inspired by Epic Games' *The Matrix Awakens: An Unreal Engine 5 Experience*—optimized for mid-range to high-end PC hardware.

1. UE5 Next-Gen Graphics Architecture (Matrix Awakens Tech Stack)

To render vast terminal structures, complex high-poly aircraft, and detailed global environments smoothly on standard gaming rigs, configure Unreal Engine 5's core rendering pipeline as follows:

A. Nanite Virtualized Geometry Configuration

- **Enable Nanite globally:** Ensure Project Settings > Rendering > `Nanite` is enabled. Convert all architectural meshes, terminal props, exterior terrain assets, and high-detail airplane exteriors/interiors to Nanite.
- **Fallback Mesh Optimization:** Set `Fallback Relative Error` to `1.0` for aircraft and terminal structures to preserve high performance when running on lower-tier GPUs.
- **Nanite Trim / Deformable Surfaces:** Keep deforming geometry (such as animated passenger characters or flex-wing mesh deformers) as standard Skeletal Meshes, utilizing Nanite strictly for static structures and rigid flight bodies.

B. Lumen Global Illumination & Reflections

- **Dynamic Lighting Engine:** Set Dynamic Global Illumination Method and Reflections to **Lumen** in Post Process Volumes and Project Settings.
- **Hardware Ray Tracing (HWRT) Scalability:**

```
r.Lumen.HardwareRayTracing 0 (Fallback to Software Ray Tracing / Distance Fields for Mid-Tier GPUs)
r.Lumen.TraceMeshSdf.Allow 1
r.Lumen.ScreenProbeGather.RadianceCache.GridResolution 48
```

- **Emissive Terminal Displays:** Utilize Lumen Emissive light propagation for flight status boards, security scanners, and overhead runway lights without placing performance-heavy dynamic lights.

C. Virtual Shadow Maps (VSM) & TSR Tuning

- **Virtual Shadow Maps:** Enable VSM in Project Settings (Shadow Map Method: `Virtual Shadow Maps`). This provides sub-pixel shadow fidelity for dense terminal structures and sun-casting airplane shadows across runways.
- **Temporal Super Resolution (TSR):** Enable TSR to render the game internally at 67% or 75% resolution (e.g., 1080p target upscaled to 1440p/4K) for substantial frame rate gains while maintaining crisp geometric clarity.

Feature	High-End PC Config	Mid-Tier PC Config (Recommended Default)
Lumen Ray Tracing	Hardware Ray Tracing (HWRT)	Software Ray Tracing (Mesh Distance Fields)
Super Resolution	TSR Quality Mode (100% Native scale)	TSR Performance / Balanced (67% Render Scale)
Virtual Shadow Maps	High Resolution Page Pool (2048 MB)	Standard Page Pool (1024 MB)
Global Foliage & Ground	Nanite Dense Foliage + Sub-surface	Nanite Standard Instanced Foliage

2. World Partition, Global Scale & Dynamic Weather

Simulating the entire world alongside microscopic airport terminal details requires Large World Coordinates (LWC) and smart spatial streaming.

A. World Partition & Large World Coordinates (LWC)

- **64-Bit LWC:** UE5 natively supports LWC. Ensure all world positioning vector math in C++ / Blueprints uses double-precision floats to prevent floating-point jitter during long-haul flights across global coordinates.
- **Grid Cell Streaming:** Configure World Partition with hierarchical streaming cells:
  - `Airport Inner Cell Size:` 128m (High detail, interior props, security systems).
  - `Airport Runway & Perimeter Cell Size:` 512m (Medium detail, ground support vehicles).
  - `Global Terrain Cell Size:` 2048m (HLOD terrain, simplified landscape).
- **Hierarchical Level of Detail (HLOD):** Build HLOD clusters for airports so whole terminals collapse into simplified Nanite proxies when viewing the airport from high altitudes during approach or departure.

B. Real-Time Environment & Global Weather System

- **Sky Atmosphere & Volumetric Clouds:** Pair the `Sky Atmosphere` component with `Volumetric Cloud` blueprints. Synchronize cloud coverage, rain, lightning, and wind vectors with live weather API feeds or real-world time-of-day tables.
- **Runway Surface Dynamics:** Map wetness/icing roughness parameters in the master runway material. Use local volumetric fog for zero-visibility instrument landings (ILS CAT III scenarios).

3. Mass Entity AI Engine (Terminal Passenger Crowds)

Similar to the city traffic and pedestrian system in *The Matrix Awakens*, **Airport Dynasty Pro** utilizes UE5's **Mass Entity Engine** (Data-Oriented ECS framework) to drive thousands of individual airport passengers smoothly.

> **Mass AI Framework Architecture**
> Standard Actor Blueprints fail when processing 500+ passengers simultaneously. The Mass Entity framework decouples data from logic, processing crowd movement, queueing, and flight boarding in cache-friendly parallel arrays.

Passenger Life-Cycle & State Machine

1. **Arrival & Entry:** Passengers spawn via public bus systems, rental car drop-offs, or parking structures based on flight departure timetables.
2. **Ticketing & Bag Drop:** Entities navigate via MassZoneGraph to kiosk lines, queueing dynamically based on airline assignment.
3. **Security Screening Routine:** Passengers approach luggage scanners, detach their luggage item entity onto conveyor belts, walk through metal detectors, and retrieve items on the opposite side.
4. **Concourse & Amenities:** Passengers evaluate dynamic Smart Objects (food courts, duty-free shops, seating areas) while waiting for boarding calls.
5. **Boarding & Deboarding:** Entities stream through jet bridges into aircraft seating nodes. Deboarding passenger entities exit through baggage claim to bus or rental car centers.

4. Real-World Flight Operations & Turnaround Logistics

Aircraft operations run on precise, real-world time frames matching actual commercial aviation workflows.

A. Turnaround Operational Workflow Table

Phase	Target Time (Narrowbody / Widebody)	Subsystem Trigger & Ground Services
Touchdown & Taxi	5–12 Mins	ATC route guidance, runway exit, gate docking system (VDGS) alignment.
Deboarding	12–25 Mins	Jetbridge / stair truck attach, door opening, Mass Entity deboarding flow start.
Catering & Refueling	20–45 Mins	Fuel truck connection, fuel pump flow calculation, galley replenishment trucks.
Baggage Handling	15–30 Mins	Baggage belt loader attach, luggage container cart transfer to sorting facilities.
Boarding & Pushback	20–35 Mins	Passenger gate call, Mass Entity seat assignment, tug boat connection & pushback clearance.

5. Interactive In-Game 3D Office Monitors

Airport management offices feature interactive 3D desktop monitors. Clicking a physical screen transitions the view cleanly into a full-screen interactive UI modal.

Step-by-Step Implementation Architecture

```
// 1. Render Target Setup on 3D Monitor Actor
UMaterialInstanceDynamic* DynamicMat = UMaterialInstanceDynamic::Create(BaseMonitorMaterial,
this);
UUserWidget* MonitorUIWidget = CreateWidget<UUserWidget>(GetWorld(), MonitorUIClass);
UWidgetRenderer* WidgetRenderer = new UWidgetRenderer(true, false);

UTextureRenderTarget2D* RenderTarget = NewObject<UTextureRenderTarget2D>(this);
RenderTarget->InitCustomFormat(1920, 1080, PF_B8G8R8A8, false);
WidgetRenderer->DrawWidget(RenderTarget, MonitorUIWidget->TakeWidget(), FVector2D(1920,
1080), 0.0f);
DynamicMat->SetTextureParameterValue(FName("DisplayTexture"), RenderTarget);
```

- **Step 1 (Physical 3D Display):** Apply the `RenderTarget` texture to the screen mesh material in the office environment so ambient lighting and reflections affect the screen.
- **Step 2 (Interaction Raycast):** Attach an `OnClicked` or Line Trace event to the 3D Monitor Static Mesh Component.
- **Step 3 (Focus Transition):** Upon player interaction, trigger a smooth camera interpolation (using a `Camera Shake / CineCamera` blend or a UMG screen focus transition) while displaying the high-resolution 2D HUD widget directly to the player screen.
- **Step 4 (Input Handling):** Set Input Mode to `Game and UI`, enabling seamless mouse control over airport cameras, live flight tracking maps, schedule timetables, and financial overview software.

> **Development Workflow Note (Claude AI Integration)**
> When engineering Airport Dynasty Pro alongside Claude AI, structure your logic requests by component modules (e.g., "Write a C++ Mass Entity Processor for terminal queueing" or "Create a Math Expression for aircraft weight and balance fuel consumption"). This guarantees clean code modularity and smooth Blueprint/C++ integration.